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## EFFECT OF WORK ENVIRONMENT ON THE PERFORMANCE OF NURSES IN PRIVATE HOSPITALS

Reshma Hota, Student MBA- Semester IV  
Ms Priyam Vishwakarma - Assistant Professor  
Faculty of Commerce and Management, Kalinga University, C.G

### **ABSTRACT:**

*The workplace atmosphere in the private health care facilities is critical in deciding the effectiveness of operations, job satisfaction, and performance of the nursing professionals. The research paper explores how the different dimensions of work environment, including physical safety and cleanliness to leadership support and teamwork, influence day-to-day performance of nurses in the private hospitals. Primary data was gathered through the use of a descriptive and analytical research design where a sample of 101 nurses were used. To analyze the existing conditions and their direct effect on accomplishing the tasks, we assessed the key variables with the help of One-Sample t-tests, Pearson Correlation, and Multiple Regression analysis.*

*We find that in most cases, the nurses view their work environment as positive (the mean scores are way beyond the neutral point), but not all of the environmental factors have an equal effect on the performance of the nurses. Although leadership support and teamwork have a positive association with performance, physical safety proved to be the most important and most significant predictor of whether or not a nurse can perform the tasks that he or she is assigned to complete successfully during a shift. The current paper can offer practical recommendations to hospital leaders who want to maximize the efficiency of nursing work by investing in workplace safety and supportive elements strategically and eventually improve the quality of patient care.*

**Keywords:** Work Environment, Nurse Performance, Private Hospitals, Physical Safety, Teamwork, Healthcare Management.

## 1. INTRODUCTION

Health care is a highly stressful and high-stakes industry, and quality of patient care is inevitably connected to the performance of frontline healthcare workers, who are mostly nurses. Within the last ten years, the pressure on nursing professionals has increased with the increase in patient loads, complicated medical equipment, and the long-term effects of world-health disasters. In private hospitals, where the operational efficiency and patient satisfaction are considered key measures of organizational success, maintaining the high level of performance of nurses is one of the main managerial goals.

Nevertheless, performance is not the product of personal ability or medical education only, but rather a strong part of the occupational ecosystem. Nurses are empowered by a supportive work environment, and fatigue and burnout are hastened by a toxic or physically hazardous work environment. The study aims at examining the particular aspects of the work environment, including physical infrastructure (lights, Ventilation, cleanliness), the manageability of workload, leadership support, teamwork as well as physical safety and the ways in which they determine performance levels of nurses.

The study aims to evaluate the current work environment and performance rates of nurses in privately owned hospitals, the correlation of the environmental variables with performance, and the most prominent predictors of an effective accomplishment of the task. With these dynamics, healthcare administrators can develop specific interventions to lower turnover, decrease medical errors, and develop a sustainable and high-performing nursing workforce.

## 2. LITERATURE REVIEW AND RESEARCH GAP

### 2.1 Healthcare Work Environments.

Healthcare work environment involves both tangible and intangible aspects, such as physical infrastructure, distribution of work load, organizational policies, style of leadership, and psychological safety. Available literature reiterates the notion that a healthcare setting that is adequately staffed, safe to work in, and whose management is both supportive is linked to better employee outcomes, especially the increased job satisfaction and improved performance. This relationship is supported by empirical evidence. The large-scale nurse data analysis demonstrated that a high patient experience score and low rates of patient safety incidents, including infections and falls, were observed in hospitals with high work environment scores (Press Ganey, 2024). Likewise, a massive systematic review and meta-analysis of evidence based on data of many hospitals showed that supportive work conditions considerably decrease burnout and enhance patient safety culture, and the role of organizational context in influencing nursing outcomes is in the center point (Lake et al., 2019).

## 2.2 Physical and Organizational Factors.

The physical and organizational factors of the work setting are very important in deciding the efficiency and clinical performance of nurses. Good lighting, air-conditioning, ergonomic workplace set-up, and access to medical equipment have been found to improve accuracy, and minimise physical stress among nurses. The level of performance is also affected by organizational issues like scheduling of shifts, balance of workload, and job description. There exists a large amount of evidence that work overload and working long hours are always linked to fatigue, stress and diminished clinical performance. Nursing workloads caused by poor staffing and extended work hours have been associated with the increased rates of errors, the reduced patient safety, and the increased burnout rates (Al Sabei et al., 2022). Human factor engineering Workload intensity that is determined by the level of staffing and work system design is a direct contributor to cognitive overload and performance decline when too high (Holden et al., 2011). Also, studies that investigate shifts as characteristics show that long working hours, night shifts, and increased fatigue are significantly correlated with each other and negatively influence the alertness of nurses and patient care quality (Booker et al., 2023).

## 2.3 Leadership and Management Support

It is well known that leadership and management support are important determinants of the work performance of nurses. Effective communication, motivation and support are components of transformational leadership styles that have been found to improve the professional commitment, job satisfaction, and overall performance of nurses. Conversely, inadequate supervision, inefficient leadership styles and recognition are factors that lead to disengagement, dissatisfaction and work ineffectiveness. A concept analysis of leadership styles in nursing stated that transformational leadership has a positive relationship with enhanced job satisfaction and less adverse patient outcomes and greater empowerment and organizational commitment among nurses (Fischer, 2016). Subsequent empirical research supports the idea that transformational leadership produces greater motivation and high performance rates than other models of leadership, and other advantages to nurse retention and quality of patient care (Boamah et al., 2018).

## 2.4 Teamwork and Interpersonal Relationships.

Teamwork and interpersonal relations are crucial elements of the healthcare working environment, as they play a significant role in the performance of nurses and patient outcomes. Teamwork helps in collaboration, coordination of care and minimizes the chances of clinical errors. Research continues to point out that friendly communication, trust, and collegiality among health practitioners positively affect job satisfaction and job performance among nurses. Effective communication, as well as teamwork among

nursing staff, are strongly positively correlated with improved team cohesion and patient care delivery, which is evidenced by empirical research (Kalisch et al., 2010). In addition, teamwork can serve as a crucial organizational asset helping nurses to address job requirements more efficiently, which contributes to improving the conditions of overall work performance and the quality of care (Van Bogaert et al., 2013).

## 2.5 Nurse Performance Outcomes

Some indicators that are normally assessed to determine nurse performance include quality of care to patients, compliance to clinical procedures, responsiveness, collaboration, and productivity. An expanding amount of empirical data show that there is a strong relationship between favourable work conditions and increased levels of nurse performance, as well as decreased burnout and better patient satisfaction. An extensive overview of literature indicated that nurse burnout has a strong correlation with worse patient safety outcomes, such as higher infection, fall, and medication error rates; worse patient satisfaction and nurse-reported quality of care (Dall'Ora et al., 2020). On the other hand, positive working conditions do reduce the adverse impact of burnout on such critical outcomes as mortality rates and length of stay, and the coordinated connection between environmental support, nurse well-being, and performance (Lake et al., 2019).

## 2.6 Gap in the Research and Objectives.

Although the literature on the work environment in healthcare is expanding, very little empirical research is directly on the topic of private hospitals. Available literature reveals lack of integration of physical, psychological and organizational variables in the study of the work environment of nurses. Moreover, there are no well-organized and holistic performance evaluation systems which directly correlate the variables of work environment with the performance outcomes of nurses. These gaps suggest that there is a need to conduct targeted research that critically looks at how various work environment dimensions affect the performance of nurses in the context of the private hospital environment.

## 3. RESEARCH DESIGN AND MODEL

For this study, a descriptive and analytical research design was adopted.

### Research Objectives

1. To assess the existing work environment of nurses in private hospitals.
2. To evaluate the level of performance among nurses in private hospitals.
3. To analyse the relationship between work environment factors and nurse performance.
4. To identify key work environment factors influencing nurse performance.

**Statement of the Problem:** The working conditions of the nurses in the private health facilities are poor and unsafe which directly affects the performance of the nurses. This results in partial shift work, hasty patient attention and high levels of stress. The hospital management will not be able to allocate resources efficiently to enhance staff efficiency and patient outcomes without determining which particular environment factors (e.g., leadership, teamwork, physical safety) have the greatest impact on the completion of tasks.

**Data Collection and Variables:**

Primary data was collected using a structured 5-point Likert scale questionnaire from a sample of 101 nurses working in private hospitals.

- **Independent Variables:** Physical Environment (Lighting/Ventilation, Cleanliness), Workload manageability, Leadership/Supervisor Support, Teamwork, and Physical Safety.
- **Dependent Variable:** Nurse Performance (measured by the ability to complete assigned tasks within the shift).

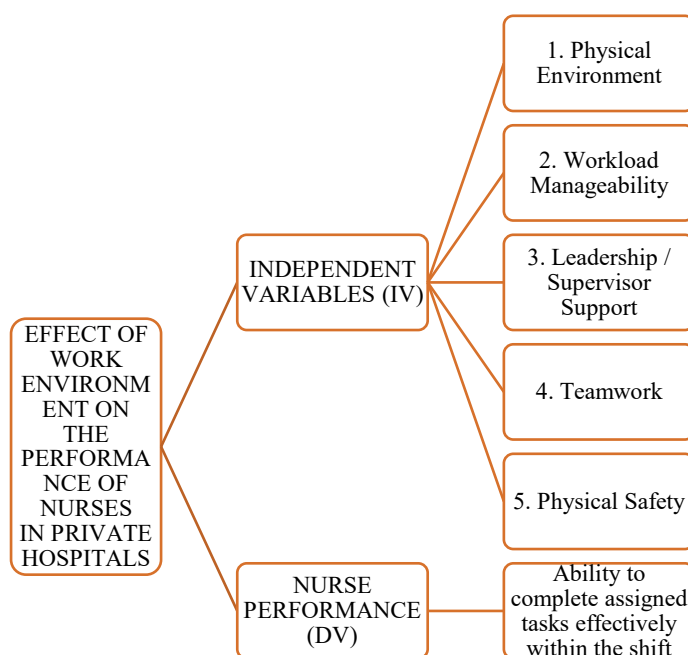


Fig no:3.1 Conceptual framework chart

**Analytical Tools:**

Data was analyzed using IBM SPSS. Techniques included Descriptive Statistics, One-Sample t-tests (to measure baseline perceptions against a neutral midpoint of 3), Pearson Correlation Analysis (to test relationships), and Multiple Regression Analysis (to identify key predictors).

#### 4. DATA ANALYSIS AND FINDINGS

Table no 4.1: Descriptives

| Descriptive Statistics                                        |     |         |         |      |                |
|---------------------------------------------------------------|-----|---------|---------|------|----------------|
|                                                               | N   | Minimum | Maximum | Mean | Std. Deviation |
| Informed_Consent_Code                                         | 101 | 1       | 1       | 1.00 | .000           |
| Age_Group_Code                                                | 101 | 1       | 5       | 2.72 | 1.517          |
| Gender Code                                                   | 101 | 1       | 3       | 1.61 | .509           |
| Highest_Nursing_Qualification_Code                            | 101 | 1       | 4       | 2.47 | 1.073          |
| Total_Years_of_Nursing_Experience_Code                        | 101 | 1       | 5       | 3.10 | 1.425          |
| Employment_Type_Code                                          | 100 | 1       | 3       | 1.27 | .584           |
| Average_Working_Hours_Per_Week_Code                           | 101 | 1       | 4       | 2.45 | 1.118          |
| Physical_Environment_Lighting_and_Ventilation_Code            | 101 | 1       | 5       | 3.56 | 1.014          |
| Physical_Environment_Clean_and_Hygienic_Unit_Code             | 101 | 1       | 5       | 3.65 | .793           |
| Physical_Environment_Layout_Supports_Safe_Movement_Code       | 101 | 1       | 5       | 3.74 | .868           |
| Physical_Environment_Sufficient_Space_for_Procedures_Code     | 101 | 1       | 5       | 3.55 | .818           |
| Physical_Environment_Noise_and_Crowding_Do_Not_Interfere_Code | 101 | 1       | 5       | 3.67 | .873           |
| Physical_Environment_Adequate_Restrooms_and_Rest_Areas_Code   | 101 | 1       | 5       | 3.62 | .881           |
| Workload_Complete_Assigned_Tasks_Within_Shift_Code            | 101 | 1       | 5       | 3.65 | .953           |
| Workload_Manageable_on_Most_Working_Days_Code                 | 101 | 1       | 5       | 3.64 | .855           |
| Workload_Frequently_Feels_Rushed_During_Patient_Care_Code     | 101 | 1       | 5       | 3.64 | .844           |

|                                                             |     |   |   |      |      |
|-------------------------------------------------------------|-----|---|---|------|------|
| Workload_Often_Misses_Scheduled_Breaks_Code                 | 101 | 1 | 5 | 3.65 | .921 |
| Workload_Documentation_Requirements_Are_Manageable_Code     | 101 | 1 | 5 | 3.76 | .777 |
| Workload_Overtime_or_Extended_Duty_is_Common_Code           | 101 | 1 | 5 | 3.73 | .882 |
| Leadership_Supervisor_Provides_Support_Code                 | 101 | 1 | 5 | 3.71 | .829 |
| Leadership_Management_Listens_to_Concerns_Code              | 101 | 1 | 5 | 3.62 | .835 |
| Leadership_Decisions_Are_Fair_and_Transparent_Code          | 101 | 1 | 5 | 3.58 | .803 |
| Leadership_Supervisor_Provides_Clear_Guidance_Code          | 101 | 1 | 5 | 3.79 | .828 |
| Leadership_Fair_and_Transparent_Decisions_Second_Item_Code  | 101 | 1 | 5 | 3.67 | .861 |
| Leadership_Training_and_Mentoring_Encouraged_Code           | 101 | 1 | 5 | 3.74 | .844 |
| Teamwork_Nurses_Cooperate_and_Support_One_Another_Code      | 101 | 1 | 5 | 3.81 | .845 |
| Teamwork_Mutual_Respect_Between_Nurses_and_Doctors_Code     | 101 | 1 | 5 | 3.69 | .880 |
| Teamwork_Conflicts_Addressed_Professionally_and_Fairly_Code | 101 | 1 | 5 | 3.72 | .826 |
| Teamwork_Communication_Effective_During_Patient_Care_Code   | 101 | 1 | 5 | 3.64 | .807 |
| Teamwork_Improves_Care_Quality_and_Reduces_Errors_Code      | 101 | 1 | 5 | 3.82 | .876 |
| Teamwork_Enough_Nurses_to_Provide_Quality_Care_Code         | 101 | 1 | 5 | 3.90 | .878 |
| Safety_Feels_Physically_Safe_in_Hospital_Code               | 101 | 1 | 5 | 3.58 | .863 |

|                                                          |     |   |   |      |      |
|----------------------------------------------------------|-----|---|---|------|------|
| Safety_Hospital_Follows_Infection_Control_Protocols_Code | 101 | 1 | 5 | 3.82 | .841 |
| Safety_Access_to_Necessary_PPE_Code                      | 101 | 1 | 5 | 3.74 | .808 |
| Safety_Protected_from_Verbal_and_Physical_Abuse_Code     | 101 | 1 | 5 | 3.73 | .859 |
| Safety_Can_Report_Mistakes_Without_Fear_of_Blame_Code    | 101 | 1 | 5 | 3.77 | .847 |
| Safety_Hospital_Supports_Mental_Wellbeing_Code           | 101 | 1 | 5 | 3.65 | .932 |
| Valid N (listwise)                                       | 100 |   |   |      |      |

The t-test was one-sample as it was determined that the perceptions of nurses about the working environment were significantly different than the neutral. The mean scores of all the chosen work environment variables, i.e., lighting and ventilation, cleanliness and hygiene, supervisor support, team work, and physical safety, were above 3 and the p-values of 0.000 were highly significant. This means that the work environment of the respondents in the private hospitals is perceived by them with a positive outlook as opposed to a neutral one. Thus, the null hypothesis is rejected, and it can be concluded that the current work environment is considered to be highly favourable by nurses.

#### Test 1

#### One-Sample t-test

#### Objective linked

To assess the existing work environment of nurses in private hospitals.

#### Hypothesis framework

- H0: The mean score of the selected work environment factors is not significantly different from the neutral level.
- H1: The mean score of the selected work environment factors is significantly different from the neutral level.

#### Basis for developing this hypothesis

This hypothesis was framed because the objective is to assess whether nurses perceive their work environment positively or negatively compared to the neutral midpoint of the scale.



Variables taken

- Physical\_Environment\_Lighting\_and\_Ventilation\_Code
- Physical\_Environment\_Clean\_and\_Hygienic\_Unit\_Code
- Leadership\_Supervisor\_Provides\_Support\_Code
- Teamwork\_Nurses\_Cooperate\_and\_Support\_One\_Another\_Code
- Safety\_Feels\_Physically\_Safe\_in\_Hospital\_Code

Why this test is suitable

This test is suitable because these are Likert-scale work environment items, and a one-sample t-test helps check whether the average response differs significantly from the neutral value 3.

Table no 4.2T-Test

| One-Sample Test                                        |                |     |                 |                 |                                           |       |
|--------------------------------------------------------|----------------|-----|-----------------|-----------------|-------------------------------------------|-------|
|                                                        | Test Value = 0 |     |                 |                 |                                           |       |
|                                                        | t              | df  | Sig. (2-tailed) | Mean Difference | 95% Confidence Interval of the Difference |       |
|                                                        |                |     |                 |                 | Lower                                     | Upper |
| Physical_Environment_Lighting_and_Ventilation_Code     | 35.325         | 100 | .000            | 3.564           | 3.36                                      | 3.76  |
| Physical_Environment_Clean_and_Hygienic_Unit_Code      | 46.306         | 100 | .000            | 3.653           | 3.50                                      | 3.81  |
| Leadership_Supervisor_Provides_Support_Code            | 45.027         | 100 | .000            | 3.713           | 3.55                                      | 3.88  |
| Teamwork_Nurses_Cooperate_and_Support_One_Another_Code | 45.329         | 100 | .000            | 3.812           | 3.65                                      | 3.98  |
| Safety_Feels_Physically_Safe_in_Hospital_Code          | 41.722         | 100 | .000            | 3.584           | 3.41                                      | 3.75  |

| One-Sample Effect Sizes                            |           |              |                |                         |       |
|----------------------------------------------------|-----------|--------------|----------------|-------------------------|-------|
|                                                    |           | Standardized | Point Estimate | 95% Confidence Interval |       |
|                                                    |           |              |                | Lower                   | Upper |
| Physical_Environment_Lighting_and_Ventilation_Code | Cohen's d | 1.014        | 3.515          | 2.989                   | 4.037 |

|                                                                                                                                                                                                       |                    |       |       |       |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|-------|-------|-------|
|                                                                                                                                                                                                       | Hedges' correction | 1.022 | 3.488 | 2.967 | 4.007 |
| Physical_Environment_Clean_and_Hygienic_Unit_Code                                                                                                                                                     | Cohen's d          | .793  | 4.608 | 3.939 | 5.273 |
|                                                                                                                                                                                                       | Hedges' correction | .799  | 4.573 | 3.910 | 5.233 |
| Leadership_Supervisor_Provides_Support_Code                                                                                                                                                           | Cohen's d          | .829  | 4.480 | 3.829 | 5.129 |
|                                                                                                                                                                                                       | Hedges' correction | .835  | 4.447 | 3.800 | 5.090 |
| Teamwork_Nurses_Cooperate_and_Support_One_Another_Code                                                                                                                                                | Cohen's d          | .845  | 4.510 | 3.855 | 5.163 |
|                                                                                                                                                                                                       | Hedges' correction | .852  | 4.476 | 3.826 | 5.124 |
| Safety_Feels_Physically_Safe_in_Hospital_Code                                                                                                                                                         | Cohen's d          | .863  | 4.152 | 3.543 | 4.757 |
|                                                                                                                                                                                                       | Hedges' correction | .870  | 4.120 | 3.517 | 4.721 |
| <p>a. The denominator used in estimating the effect sizes.<br/>Cohen's d uses the sample standard deviation.<br/>Hedges' correction uses the sample standard deviation, plus a correction factor.</p> |                    |       |       |       |       |

The one-sample t-test was employed to determine the significant difference between the perception of the work environment by the nurses and the neutral level. The mean scores of all the chosen work environment variables, i.e., lighting and ventilation, cleanliness and hygiene, supervisor support, teamwork, and physical safety were above 3 and had a very high p-value of 0.000. This implies that the respondents tend to have a positive attitude towards the work environment in any private hospital as opposed to a neutral one. Thus, the null hypothesis is rejected and it is determined that the current working environment is considered much more preferable by nurses.

## Test 2

### One-Sample t-test

### Objective linked

To evaluate the level of performance among nurses in private hospitals.

### Hypothesis framework

- H0: The mean score of nurse performance is not significantly different from the neutral level.
- H1: The mean score of nurse performance is significantly different from the neutral level.

Basis for developing this hypothesis

This hypothesis was framed because the objective is to determine whether the reported level of nurse performance is meaningfully above or below the midpoint of the scale.

Variables taken

- Workload\_Complete\_Assigned\_Tasks\_Within\_Shift\_Code

Why this test is suitable

This test is suitable because this is the most direct performance-related item in the SPSS-ready sheet, and one-sample t-test helps assess whether the average performance response differs from neutral.

Table no 4.3 One-Sample Test

| One-Sample Test                                    |                |     |                 |                 |                                           |       |
|----------------------------------------------------|----------------|-----|-----------------|-----------------|-------------------------------------------|-------|
|                                                    | Test Value = 0 |     |                 |                 |                                           |       |
|                                                    | t              | df  | Sig. (2-tailed) | Mean Difference | 95% Confidence Interval of the Difference |       |
|                                                    |                |     |                 |                 | Lower                                     | Upper |
| Workload_Complete_Assigned_Tasks_Within_Shift_Code | 38.517         | 100 | .000            | 3.653           | 3.47                                      | 3.84  |

Table no 4.4 One-Sample Effect Sizes

| One-Sample Effect Sizes                                                                                                                                                                               |                    |              |                |                         |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------|----------------|-------------------------|-------|
|                                                                                                                                                                                                       |                    | Standardized | Point Estimate | 95% Confidence Interval |       |
|                                                                                                                                                                                                       |                    |              |                | Lower                   | Upper |
| Workload_Complete_Assigned_Tasks_Within_Shift_Code                                                                                                                                                    | Cohen's d          | .953         | 3.833          | 3.266                   | 4.396 |
|                                                                                                                                                                                                       | Hedges' correction | .960         | 3.804          | 3.241                   | 4.363 |
| <p>a. The denominator used in estimating the effect sizes.<br/>Cohen's d uses the sample standard deviation.<br/>Hedges' correction uses the sample standard deviation, plus a correction factor.</p> |                    |              |                |                         |       |

The one-sample t-test was also used to assess the degree of nurse performance based on the variable of accomplishing the assigned duties within the shift time. The average is 3.65, which is above the midpoint or the neutral, and the value is statistically significant at  $p = 0.000$ . This reveals that nurses have a general consensus on the fact that they can accomplish their roles assigned to them in their shift. Hence, the null hypothesis is rejected and it is concluded that the level of nurse performance is significantly above the average neutral level.

### Test 3

#### Pearson Correlation

#### Objective linked

To analyse the relationship between work environment factors and nurse performance.

#### Hypothesis framework

- H0: There is no significant relationship between work environment factors and nurse performance.
- H1: There is a significant relationship between work environment factors and nurse performance.

#### Basis for developing this hypothesis

This hypothesis was framed because the objective specifically examines whether better work environment conditions are associated with better nurse performance.

#### Variables taken

- Performance variable:
  - Workload\_Complete\_Assigned\_Tasks\_Within\_Shift\_Code
- Work environment variables:
  - Physical\_Environment\_Lighting\_and\_Ventilation\_Code
  - Workload\_Manageable\_on\_Most\_Working\_Days\_Code
  - Leadership\_Supervisor\_Provides\_Support\_Code
  - Teamwork\_Nurses\_Cooperate\_and\_Support\_One\_Another\_Code
  - Safety\_Feels\_Physically\_Safe\_in\_Hospital\_Code

The test is appropriate since these are coded scale variables and Pearson correlation assists in the strength and direction of the relationship between the environmental factors of work and nurse performance.

Table no: 4.5 Correlations

| Correlations                                       |                         |                                                    |                                                    |                                               |                                             |                                                        |                                               |
|----------------------------------------------------|-------------------------|----------------------------------------------------|----------------------------------------------------|-----------------------------------------------|---------------------------------------------|--------------------------------------------------------|-----------------------------------------------|
|                                                    |                         | Workload_Complete_Assigned_Tasks_Within_Shift_Code | Physical_Environment_Lighting_and_Ventilation_Code | Workload_Manageable_on_Most_Working_Days_Code | Leadership_Supervisor_Provides_Support_Code | Teamwork_Nurses_Cooperate_and_Support_One_Another_Code | Safety_Feels_Physically_Safe_in_Hospital_Code |
| Workload_Complete_Assigned_Tasks_Within_Shift_Code | Pearson Correlation     | 1                                                  | .142                                               | .178                                          | .253*                                       | .303**                                                 | .394**                                        |
|                                                    | Significance (2-tailed) |                                                    | .156                                               | .075                                          | .011                                        | .002                                                   | .000                                          |

|                                                    |                     |      |       |       |        |        |       |
|----------------------------------------------------|---------------------|------|-------|-------|--------|--------|-------|
|                                                    | N                   | 101  | 101   | 101   | 101    | 101    | 101   |
| Physical_Environment_Lighting_and_Ventilation_Code | Pearson Correlation | .142 | 1     | .223* | .349** | .183   | .157  |
|                                                    | Sig. (2-tailed)     | .156 |       | .025  | .000   | .066   | .118  |
|                                                    | N                   | 101  | 101   | 101   | 101    | 101    | 101   |
| Workload_Manageable_on_Most_Working_Days_Code      | Pearson Correlation | .178 | .223* | 1     | .334** | .266** | .217* |

|                                                         |                                                                |       |        |        |      |        |        |
|---------------------------------------------------------|----------------------------------------------------------------|-------|--------|--------|------|--------|--------|
|                                                         | o<br>n                                                         |       |        |        |      |        |        |
|                                                         | Si<br>g.<br>(2<br>-<br>ta<br>il<br>ed<br>)                     | .075  | .025   |        | .001 | .007   | .029   |
|                                                         | N                                                              | 101   | 101    | 101    | 101  | 101    | 101    |
| Leadership_S<br>upervisor_Pr<br>ovides_Supp<br>ort_Code | P<br>ea<br>rs<br>o<br>n<br>C<br>or<br>re<br>la<br>ti<br>o<br>n | .253* | .349** | .334** | 1    | .365** | .307** |
|                                                         | Si<br>g.<br>(2<br>-<br>ta<br>il<br>ed<br>)                     | .011  | .000   | .001   |      | .000   | .002   |

|                                                        |                     |        |      |        |        |        |        |
|--------------------------------------------------------|---------------------|--------|------|--------|--------|--------|--------|
|                                                        | N                   | 101    | 101  | 101    | 101    | 101    | 101    |
| Teamwork_Nurses_Cooperate_and_Support_One_Another_Code | Pearson Correlation | .303** | .183 | .266** | .365** | 1      | .262** |
|                                                        | Sig. (2-tailed)     | .002   | .066 | .007   | .000   |        | .008   |
|                                                        | N                   | 101    | 101  | 101    | 101    | 101    | 101    |
| Safety_Feels_Physically_Safe_in_Hospital_Code          | Pearson Correlation | .394** | .157 | .217*  | .307** | .262** | 1      |



|                                                              |                                            |      |      |      |      |      |     |
|--------------------------------------------------------------|--------------------------------------------|------|------|------|------|------|-----|
|                                                              | o<br>n                                     |      |      |      |      |      |     |
|                                                              | Si<br>g.<br>(2<br>-<br>ta<br>il<br>ed<br>) | .000 | .118 | .029 | .002 | .008 |     |
|                                                              | N                                          | 101  | 101  | 101  | 101  | 101  | 101 |
| *. Correlation is significant at the 0.05 level (2-tailed).  |                                            |      |      |      |      |      |     |
| **. Correlation is significant at the 0.01 level (2-tailed). |                                            |      |      |      |      |      |     |

Pearson correlation test was employed to analyze the correlation between the work environment factors and the nurse performance. The findings reveal that there are positive correlations between leadership support, teamwork, and physical safety and performance, with physical safety having the highest association ( $r = 0.394$ ,  $p = 0.000$ ). But lighting and ventilation, and workload manageable do not display statistically significant correlations with performance at the 5% level. Thus, the null hypothesis is rejected partially, and it is concluded that there is some significant work environment factors, which are positively associated with nurse performance.

Test 4

Multiple Regression

Objective linked

To identify key work environment factors influencing nurse performance.

Hypothesis framework

- H0: Work environment factors do not significantly influence nurse performance.
- H1: Work environment factors significantly influence nurse performance.

Basis for developing this hypothesis

This hypothesis was framed because the objective is to identify which work environment factors have the strongest effect on nurse performance.

Variables taken

- Dependent variable:
  - Workload\_Complete\_Assigned\_Tasks\_Within\_Shift\_Code
- Independent variables:
  - Physical\_Environment\_Lighting\_and\_Ventilation\_Code
  - Workload\_Manageable\_on\_Most\_Working\_Days\_Code
  - Leadership\_Supervisor\_Provides\_Support\_Code
  - Teamwork\_Nurses\_Cooperate\_and\_Support\_One\_Another\_Code
  - Safety\_Feels\_Physically\_Safe\_in\_Hospital\_Code

The test is appropriate as there is a dependent variable and several predictor variables and regression assists in the determination of the main work environment variables that contribute to nurse performance.

Regression

Table no: 4.6 Model Summary

| Model Summary <sup>b</sup>                                                                                                                                                                                                                                                       |                   |          |                   |                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------|-------------------|----------------------------|
| Model                                                                                                                                                                                                                                                                            | R                 | R Square | Adjusted R Square | Std. Error of the Estimate |
| 1                                                                                                                                                                                                                                                                                | .454 <sup>a</sup> | .206     | .164              | .871                       |
| a. Predictors: (Constant), Safety_Feels_Physically_Safe_in_Hospital_Code, Physical_Environment_Lighting_and_Ventilation_Code, Workload_Manageable_on_Most_Working_Days_Code, Teamwork_Nurses_Cooperate_and_Support_One_Another_Code, Leadership_Supervisor_Provides_Support_Code |                   |          |                   |                            |
| b. Dependent Variable: Workload_Complete_Assigned_Tasks_Within_Shift_Code                                                                                                                                                                                                        |                   |          |                   |                            |

Table no: 4.7 ANOVA

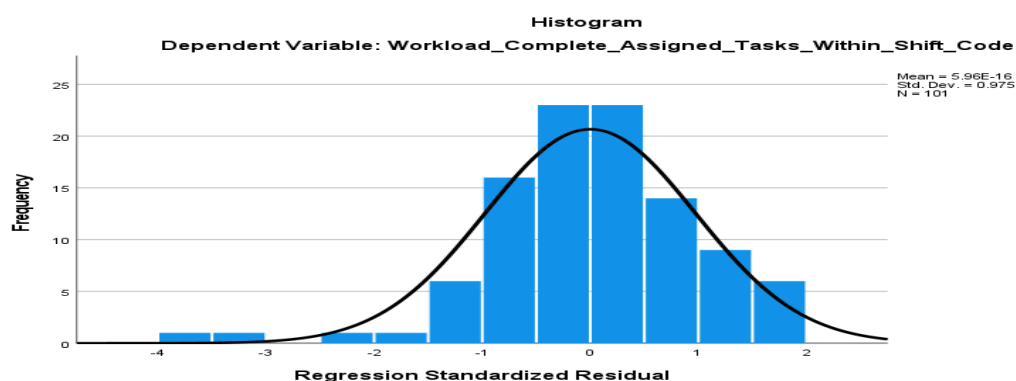
| ANOVA <sup>a</sup> |            |                |    |             |       |                   |
|--------------------|------------|----------------|----|-------------|-------|-------------------|
| Model              |            | Sum of Squares | df | Mean Square | F     | Sig.              |
| 1                  | Regression | 18.734         | 5  | 3.747       | 4.934 | .000 <sup>b</sup> |
|                    | Residual   | 72.138         | 95 | .759        |       |                   |

|                                                                                                                                                                                                                                                                                  |       |        |     |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-----|--|--|--|
|                                                                                                                                                                                                                                                                                  | Total | 90.871 | 100 |  |  |  |
| a. Dependent Variable: Workload Complete Assigned Tasks Within Shift Code                                                                                                                                                                                                        |       |        |     |  |  |  |
| b. Predictors: (Constant), Safety_Feels_Physically_Safe_in_Hospital_Code, Physical_Environment_Lighting_and_Ventilation_Code, Workload_Manageable_on_Most_Working_Days_Code, Teamwork_Nurses_Cooperate_and_Support_One_Another_Code, Leadership_Supervisor_Provides_Support_Code |       |        |     |  |  |  |

Table no: 4.8 Coefficients

| Coefficients <sup>a</sup>                                                 |                                                        |                             |            |                           |       |      |
|---------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------|------------|---------------------------|-------|------|
| Model                                                                     |                                                        | Unstandardized Coefficients |            | Standardized Coefficients | t     | Sig. |
|                                                                           |                                                        | B                           | Std. Error | Beta                      |       |      |
| 1                                                                         | (Constant)                                             | 1.107                       | .577       |                           | 1.919 | .058 |
|                                                                           | Physical_Environment_Lighting_and_Ventilation_Code     | .027                        | .093       | .028                      | .288  | .774 |
|                                                                           | Workload_Manageable_on_Most_Working_Days_Code          | .036                        | .111       | .032                      | .324  | .747 |
|                                                                           | Leadership_Supervisor_Provides_Support_Code            | .080                        | .124       | .069                      | .643  | .522 |
|                                                                           | Teamwork_Nurses_Cooperate_and_Support_One_Another_Code | .205                        | .114       | .182                      | 1.805 | .074 |
|                                                                           | Safety_Feels_Physically_Safe_in_Hospital_Code          | .347                        | .108       | .314                      | 3.203 | .002 |
| a. Dependent Variable: Workload Complete Assigned Tasks Within Shift Code |                                                        |                             |            |                           |       |      |

Charts no 4.1 Histogram



The multiple regression analysis was conducted to determine the most important work environment factors that affected nurse performance. The model is statistically significant in general,  $F = 4.934$  and  $p = 0.000$  and describes 20.6% of the variance in nurse performance. Physical safety in the hospital is the only significant predictor ( $Beta = 0.314$ ,  $p = 0.002$ ) in the combined model, whereas the other variables are not significant. Therefore, the null hypothesis is rejected and the conclusion made that factors in the work environment play an important role in nurse performance, physical safety being the most important key predictor.

## 5. LIMITATIONS OF THE STUDY

While this study provides valuable empirical insights, it is subject to certain limitations:

1. **Sample Size and Scope:** The study relies on a sample size of 101 nurses from private hospitals, which may limit the generalizability of the findings to public hospitals or rural healthcare centers where resource constraints are more severe.
2. **Self-Report Bias:** Performance was measured using self-reported capability (completing tasks within a shift) rather than objective clinical metrics (e.g., patient recovery times or administrative audits), which may introduce subjective bias.
3. **Cross-Sectional Nature:** The data represents a snapshot in time. A longitudinal study would better capture how changes in the work environment over time dynamically influence burnout and performance.

## 6. CONCLUSION AND FUTURE SCOPE

Conclusion:

This paper confirms that the workplace conditions in privately owned hospitals have a strong influence on the performance of the nurses. Although such factors as clean infrastructure, transformational leadership and peer-teamwork make the occupational atmosphere positive, they are less important than the necessity of physical safety. The empirical evidence confirms that the more physically secure and safe nurses are against occupational risks, the more productive their work and the performance of their shifts is. To hospital management, it is obvious that not only is it an ethical duty to invest in infection control, proper PPE and zero-tolerance to workplace abuse, it is also the most efficient business strategy to improve the performance of the nurses.

Future Scope:

The future study ought to elaborate this model by providing a comparative study between the state and the private healthcare sectors. Moreover, the use of objective performance reviews and patient satisfaction scores as the dependent variables would give a more comprehensive picture of the effects of the nursing working environment on the overall healthcare delivery chain. Longitudinal studies that evaluate the effects of specific safety measures on the retention rates of the nurses are also strongly suggested.

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